# Tree exercises

1. Prove that a finite connected graph is a tree iff it has n−1 edges and no cycle.
2. A full binary tree has 31 leaves. Find its internal-node and edge counts and the minimum attainable height with root depth zero.
3. Use comparison decision trees to prove that sorting six distinct keys needs at least ten comparisons in the worst case. Evaluate $\lceil \log_2 6! \rceil$. Merge-sort [5,1,4,2,6,3], splitting odd lengths with a left half of floor(n/2), and count comparisons by level. If this input uses ten, does that prove optimal worst-case comparison count on all six-key inputs?
4. Build a Huffman tree for frequencies 2,3,5,7,11. Give a prefix code, weighted cost, and mean length.
5. Prove uniqueness of simple paths in a tree using the cycle formed by two different paths.
6. Draw a connected five-vertex graph with a one-child DFS root and a nonroot cut vertex. Apply the low-link cut criterion.
7. On edges AB, BC, CA, CD, DE, EF, FD, DG, start alphabetic DFS at A. Trace disc, low, parent, edge-ID pushes, and every complete pop. Find cuts, bridges, and vertex blocks. Then root at D; explain root-child and cut-criterion changes and outputs that must remain invariant.
8. Prove prefix-code lengths (1,2,2,3) impossible, and construct words for (2,2,3,3,3,3). For one symbol, distinguish optimal length zero from a specification requiring nonempty words.
9. A full ternary tree has thirteen internal nodes. Find total nodes, leaves, edges, and minimum height with root depth zero. Prove attainability rather than only a lower bound.
10. For positive weights $x \le y \le z$, prove the minimum binary-prefix weighted cost and give optimal words. Explain why z at depth two cannot improve the result. Evaluate merge cost and mean length for 2,5,9.
11. Prove every tree edge is a bridge and $\sum_v (\deg(v) - 2) = -2$. Find leaves when three vertices have degree three, one degree five, and every other nonleaf degree two. Explain the singleton exception to degree-one leaf counting.
12. A loop-free multigraph has vertices u, v and parallel IDs $e_1$,$e_2$. DFS discovers v from u along $e_1$. Use complete low-link rules to decide whether $e_1$ is a bridge and explain the error from skipping every edge to the parent vertex.

# Tree exercises (continued)

### References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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